Bayesian Optimization Report: Function 6

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.665	Medium
2	0.665	Medium
3	0.665	Medium
4	0.666	Medium
5	0.667	Medium
6	0.669	Medium
7	0.668	Medium
8	0.669	Medium
9	0.671	Medium
10	0.671	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.2921	0.1543	0.4763
2	0.2902	0.1459	0.4407
3	0.2758	0.1027	0.4562
4	0.2709	0.1379	0.4196
5	0.2628	0.1239	0.3800
6	0.2596	0.1282	0.3789
7	0.2063	0.0984	0.4300
8	0.2043	0.0819	0.4830
9	0.2022	0.0763	0.4448
10	0.1925	0.0984	0.4677

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.45355215 0.39419141 0.61440534 0.54576678 0.]	-0.1707223077608615	Matern(ν=0.5)	EI	0.665	None

2	[0.45188367 0.98284556 0.73219382 0.36538473 0.86104907]	-1.8762251172341786	Matern(ν=0.5)	EI	0.665	None
3	[0.43356154 0.94669747 0. 0.35902395 0.8660701]	-2.2938404294540757	Matern(ν=0.5)	EI	0.665	None
4	[0.43519362 0.98404189 0.60228255 0.26740129 0.69261483]	-1.8917846722902079	Matern(ν=0.5)	EI	0.666	None
5	[0.84709754 1. 0.40659705 0.13020169 0.77344271]	-2.60925726006269	Matern(ν=0.5)	EI	0.667	None
6	[0.43660427 0.89023426 0.17053947 0.21046559 0.70115338]	-2.1439857804933324	Matern(ν=0.5)	EI	0.669	None
7	[0.48543816 0.41187979 0.49397804 0.7795048 0.]	-0.15707605361592014	Matern(ν=0.5)	EI	0.668	None
8	[0.46154139 0.87568294 0.15861816 0.23090526 0.89296299]	-2.261007636110517	Matern(ν=0.5)	EI	0.669	None
9	[1. 1. 0.73940885 0.73153613 1.]	-2.3552578537859787	Matern(ν=0.5)	EI	0.671	None
10	[0.97946 0.9755066 0.52926346 0.17888458 0.81461731]	-2.7037074858079047	Matern(ν=0.5)	EI	0.671	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.613233
EI	0.029037
PI	0.242981
THOMPSON	1.393710
KG	0.029037
ENTROPY	2.712964
PORTFOLIO	1.500368

Iteration 2

Method	Max Value
UCB	1.247303
EI	0.004957
PI	0.043981

THOMPSON	1.211649
KG	0.004957
ENTROPY	2.701369
PORTFOLIO	1.459263

Iteration 3

Method	Max Value
UCB	1.273975
EI	0.006623
PI	0.069255
THOMPSON	1.287375
KG	0.006623
ENTROPY	2.564645
PORTFOLIO	1.325419

Iteration 4

Method	Max Value
UCB	1.291621
EI	0.008002
PI	0.084112
THOMPSON	1.213554
KG	0.008002
ENTROPY	2.976999
PORTFOLIO	1.394655

Iteration 5

Method	Max Value
UCB	1.263065
EI	0.004635
PI	0.055198
THOMPSON	1.397922
KG	0.004635
ENTROPY	2.755107
PORTFOLIO	1.324526

Iteration 6

Method	Max Value
UCB	1.286033
EI	0.004739
PI	0.071976
THOMPSON	1.208589
KG	0.004739
ENTROPY	2.630917
PORTFOLIO	1.134200

Iteration 7

Method	Max Value
UCB	1.361423
EI	0.022029
PI	0.260972
THOMPSON	1.274008
KG	0.022029
ENTROPY	2.363602
PORTFOLIO	1.033610

Iteration 8

Method	Max Value
UCB	1.288982
EI	0.004601
PI	0.075304
THOMPSON	1.246984
KG	0.004601
ENTROPY	2.207095
PORTFOLIO	1.035720

Iteration 9

Method	Max Value
UCB	1.270260
EI	0.001937
PI	0.038812

THOMPSON	1.278443
KG	0.001937
ENTROPY	2.548465
PORTFOLIO	0.957960

Iteration 10

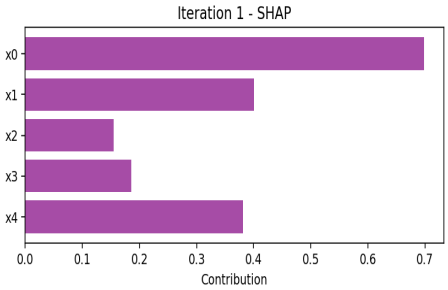
Method	Max Value
UCB	1.324532
EI	0.008888
PI	0.155150
THOMPSON	1.292938
KG	0.008888
ENTROPY	2.683822
PORTFOLIO	0.992806

LOO MSE Summary

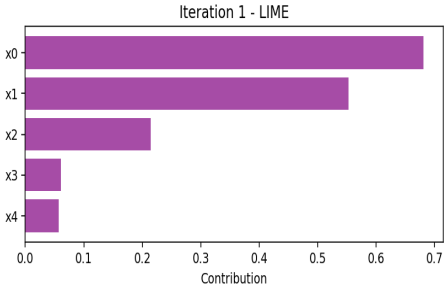
Iteration	LOO MSE
1	0.088501
2	0.128494
3	0.116794
4	0.090030
5	0.085252
6	0.079944
7	0.058733
8	0.047940
9	0.045725
10	0.040804

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



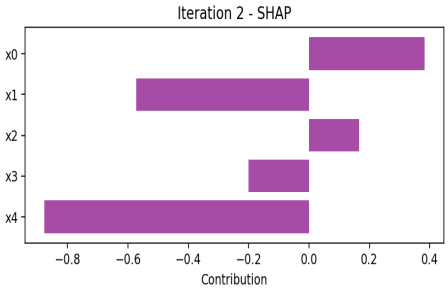
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.69812
x1	0.40082
x2	0.15506
x3	0.18522
x4	0.38147

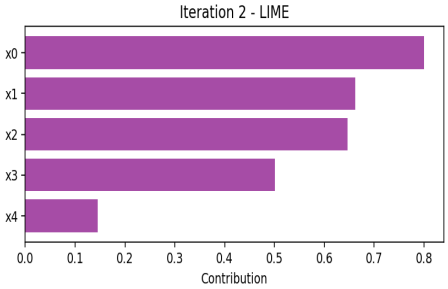
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.68093
x1	0.55315
x2	0.21413
x3	0.06037
x4	0.05685

Iteration 2 - SHAP



Iteration 2 - LIME



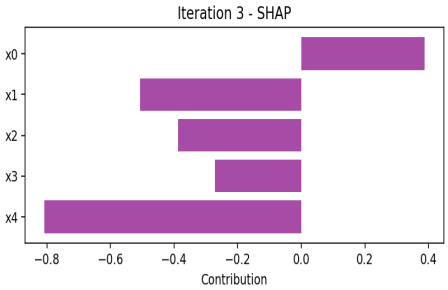
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.38368
x1	-0.57302
x2	0.16605
x3	-0.20015
x4	-0.87737

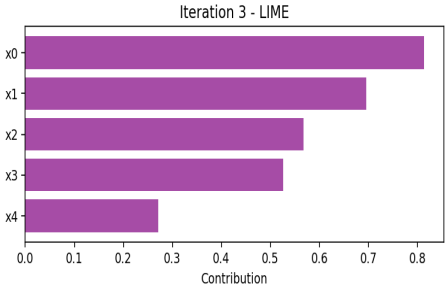
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.79986
x1	0.66238
x2	0.64655
x3	0.50042
x4	0.14476

Iteration 3 - SHAP



Iteration 3 - LIME



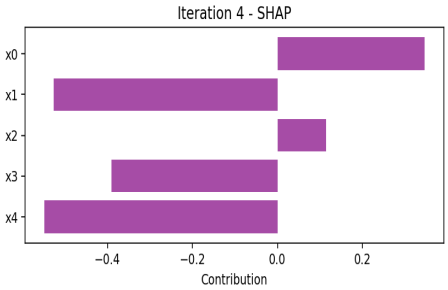
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.38835
x1	-0.50720
x2	-0.38826
x3	-0.27202
x4	-0.80824

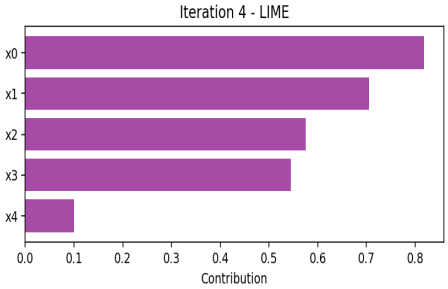
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.81336
x1	0.69552
x2	0.56769
x3	0.52659
x4	0.27179

Iteration 4 - SHAP



Iteration 4 - LIME



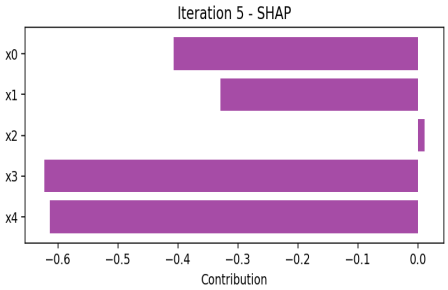
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.34700
x1	-0.52645
x2	0.11480
x3	-0.39092
x4	-0.54883

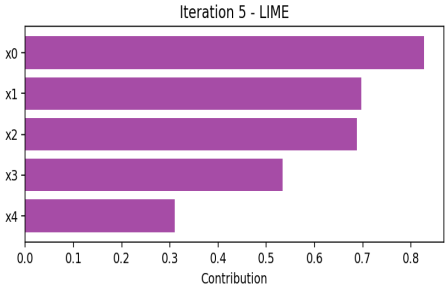
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.81775
x1	0.70572
x2	0.57545
x3	0.54452
x4	0.10031

Iteration 5 - SHAP



Iteration 5 - LIME



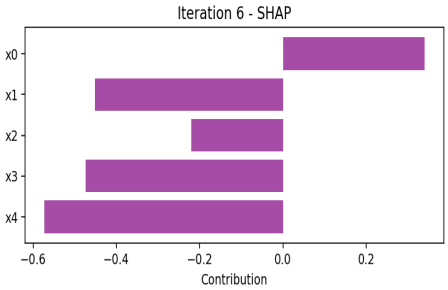
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.40675
x1	-0.32882
x2	0.01133
x3	-0.62249
x4	-0.61352

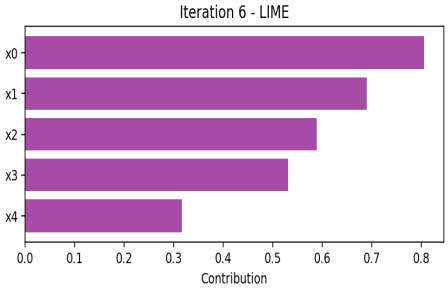
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.82695
x1	0.69665
x2	0.68794
x3	0.53423
x4	0.31038

Iteration 6 - SHAP



Iteration 6 - LIME



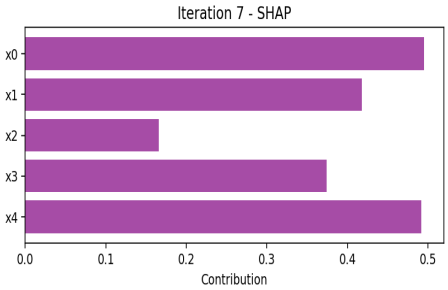
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.34082
x1	-0.45135
x2	-0.21991
x3	-0.47431
x4	-0.57301

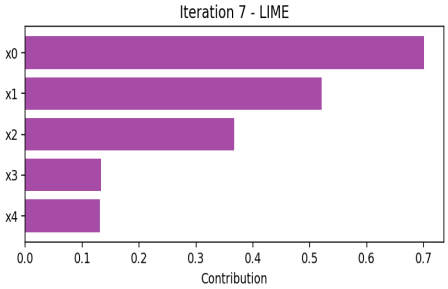
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.80503
x1	0.69012
x2	0.58836
x3	0.53123
x4	0.31645

Iteration 7 - SHAP



Iteration 7 - LIME



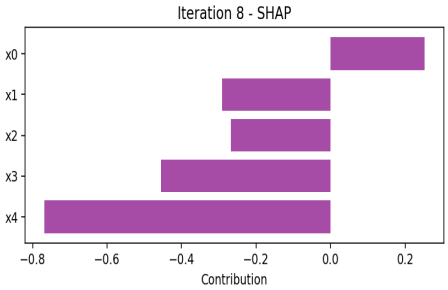
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	0.49489
x1	0.41762
x2	0.16555
x3	0.37411
x4	0.49148

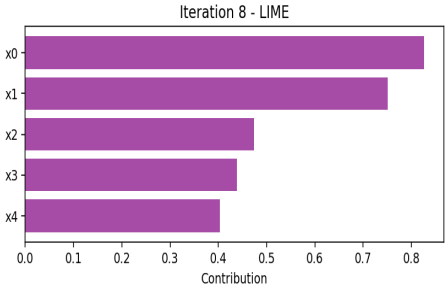
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.70076
x1	0.52096
x2	0.36697
x3	0.13283
x4	0.13155

Iteration 8 - SHAP



Iteration 8 - LIME



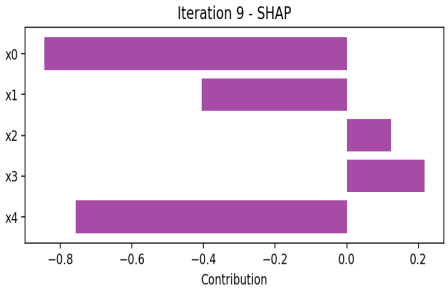
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.25256
x1	-0.29181
x2	-0.26839
x3	-0.45500
x4	-0.76831

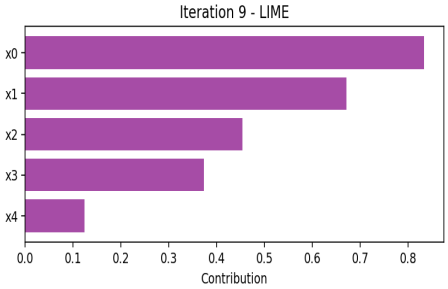
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.82594
x1	0.75056
x2	0.47383
x3	0.43865
x4	0.40320

Iteration 9 - SHAP



Iteration 9 - LIME



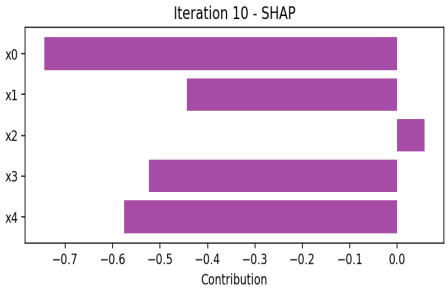
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	-0.84481
x1	-0.40451
x2	0.12391
x3	0.21807
x4	-0.75788

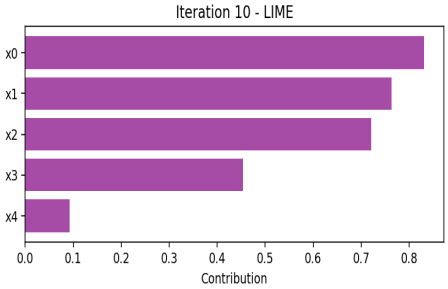
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.83281
x1	0.67073
x2	0.45335
x3	0.37272
x4	0.12350

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	-0.74397
x1	-0.44356
x2	0.05791
x3	-0.52335
x4	-0.57595

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.83069
x1	0.76331
x2	0.72129
x3	0.45311
x4	0.09287